

ARPITA PRIYADARSHANI SAMAL

Metallurgical and Materials Engineer | Casting & Solidification | ICME |

Phase Field Modelling | Thermo-Calc (CALPHAD) | Light Metal Alloys

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Education

Degree	Institute	CGPA	Years
PhD	Indian Institute of Technology Madras	9.5	2022-ongoing
M.Tech	Indian Institute of Technology Jodhpur	9.15	2020 - 22
B. Tech	Veer Surendra Sai University of Technology, Sambalpur	9.18	2016 - 20

Projects

PhD Project: Phase Field Modelling and microstructure development via ICME approach for casting and solidification of aluminium alloys

An ICME framework to model microstructure evolution during casting and solidification of aluminium alloys, using Phase Field Modelling and Thermo-Calc (CALPHAD) to predict phase diagrams, segregation behaviour, and grain morphology.

M.Tech Project: Role of microstructure and bio-corrosion behavior of Mg rich Mg-Zn-Nd alloys

The work incorporated neodymium additions (upto 5%) in Mg-4Zn alloy in order to improve mechanical and corrosion resistance property which found its application in biomedical implants.

B.Tech Project: Effect of Sintering Temperature and Reinforcement Composition on Microstructure and Mechanical properties of Al-SiC-Fly Ash Hybrid Composite using Powder Metallurgy Route

Fabricated Al-SiC-Fly Ash hybrid composites via powder metallurgy, varying sintering temperature to optimize hardness, density, and compressive strength – work published in Springer (Processing and Characterization of Materials, 2023, DOI: 10.1007/978-981-99-5509-1_8).

Achievements

- Poster presentation in the 7th **International Conference** on Processing and Characterization of Materials (ICPCM-2025), NIT Rourkela, 12nd -13th December, 2025.
- Poster presentation in the 30th **International Conference** on Processing of Advanced Materials and Fabrication of Products (PAMFP-2024), IIT Hyderabad, 11st -13th December, 2024.
- Poster presentation in the 28th **International Conference** on Non-ferrous Metals - 2024 (ICNFM - 2024), Bhubaneswar, 12nd -13th July, 2024.
- Poster presentation in the **National Symposium** - “Recent Trends in Material Science”, AMALGAM, IIT MADRAS, 3rd March, 2024.
- Paper presentation in the “2nd Virtual **National Conference** on Advancement in Materials Processing Technology” at NIT, Jamshedpur, 31st October– 2nd November, 2020.
- Paper presentation in “Webinar for Young Researchers in Metallurgical & Materials Science”, CSIR – National Metallurgical Laboratory, Jamshedpur, 12th – 14th August, 2020.
- Paper presentation in “**International Conference** on Innovations in Aluminium Technology” at IIM, Angul, 8th-9th February, 2020.
- Paper presentation in the “4th **International Conference** on Advances in Steel, Power and Construction Technology” at OPJindal University, Raigarh during 21st -22nd November, 2019.
- Poster presentation entitled in TEQIP-III sponsored “NCFAM - **National Conference** on Frontiers in Advanced Materials” at VSSUT, Burla, Odisha during 27th - 28th July, 2019.

Industrial Training and Internships

- ThermoCalc training, organised by **Bhanu Scientific Systems Pvt Ltd (BSSPL)**, Hyderabad (3rd - 6th May, 2023).
- Summer Research Training Programme, **CSIR-Institute of Minerals and Materials Technology**, Bhubaneswar (1st June – 31st August, 2020).
- Vocational training in Cast House, Smelter Plant, **Aditya Aluminium and Power Plant**, Lapanga, Sambalpur (3rd June – 3rd July, 2019).
- Research intern at **National Institute of Technology**, Rourkela (15th December–31st December, 2018).
- Vocational training in Blast Furnace, **SAIL, Rourkela Steel Plant** (28th May – 27th June, 2018).

Skills and Expertise

Metallography: Expertise on the metallography of ferrous and non-ferrous alloys.

Mechanical testing: Tensile testing, and various aspects of hardness testing.

Characterization: Optical microscopy, SEM, TEM, HRTEM, XRD, ImageJ.

Software skills: Thermo-calc, Phase field simulation, Origin Pro, C Programming and C++, Microsoft Office.

Awards

- **Silver medal** for Best Academic Performance, M.Tech Materials Engineering Program, IIT Jodhpur, 2022.
- **1st position** in Poster presentation at International Conference on Non-ferrous Metals (ICNFM), 2024, Bhubaneswar.
- **2nd position** in Poster presentation at International Conference on Processing of Advanced Materials and Fabrication of Products, 2024, IIT Hyderabad.
- **Winner** of OLA Sankalp 2024 Thought Paper Contest.
- **Winner** of Digital Clash Competition, AIRSS, IIT Madras, 2024.
- **2nd prize** in Poster Making Competition, IIT Madras, 2024.
- **1st position** in Sri Vasudevan Quiz (Meta Quiz) at Amalgam, Department of Metallurgical and Materials Engineering, IIT Madras, 2023.
- **1st position** in Online Technical Presentation at Diamond Jubilee Year Celebration, NIT Rourkela, 2021.
- **1st position** in Online Quiz Contest at Diamond Jubilee Year Celebration, NIT Rourkela, 2021.
- **2nd position** in Poster presentation at IIM, Reminiscence organised by IEST, Shibpur.

Declaration

I hereby declare that the information stated above is true to the best of my knowledge.

Dt – 04/06/2026

Arpita Priyadarshani Samal